Sec 11.7, (h.7, 8.9)

Prob.4. Find the Fourier (i) cosine & sin series of the f-n

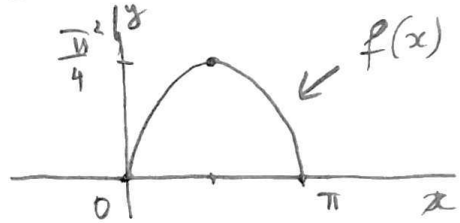
(1) $f(x) = x(\pi - x)$, on $0 \le x \le \pi$, &

then sketch the graphs of two f-ns to which the series converges for all x. You may use technology.

Sol n.

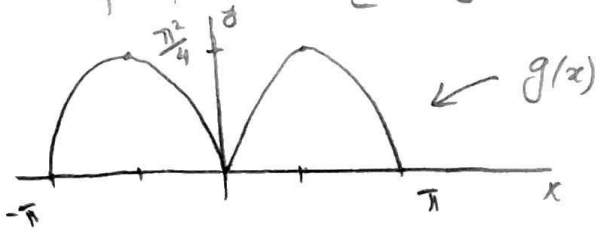
Let

(2) $f(x) = x(\pi - x)$, on $0 \le x \le \pi$.



(i) A let $g(x)$ to be even extension of $f(x)$ on $[-\pi, \pi]$:

(3) $g(x) := \begin{cases} f(x), & 0 \le x \le \pi \\ \boxed{\cdot}, & -\pi \le x < 0 \end{cases} \Rightarrow$



Then, the F-S of even f-n $g(x)$ on $[-\pi, \pi]$ is

(a) $g(x)$ is piecewise smooth
(b) $g(-\pi) = g(\pi)$

(4) F-S of $g(x) = g(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$ where

(5) $a_0 = \frac{2}{\pi} \int_0^{\pi} g(x)\, dx$

(6) $a_n = \frac{2}{\pi} \int_0^{\pi} g(x) \cos(nx)\, dx$, $n = 1, 2, \ldots$

(7) $b_n = 0$, for $n = 1, 2, \ldots$ (by the result of pr.8 in HW 10).

Then,

$$a_0 = \frac{2}{\pi} \int_0^{\pi} \underbrace{g(x)}_{= f(x) \text{ on } [0,\pi]}\, dx = \frac{2}{\pi} \int_0^{\pi} f(x)\, dx = \frac{2}{\pi} \cdot \int_0^{\pi} x(\pi - x)\, dx = \frac{2}{\pi} \int_0^{\pi} \pi x - x^2\, dx$$

$$= \frac{2}{\pi} \cdot \left[\frac{\pi x^2}{2} - \frac{x^3}{3} \right]_{x=0}^{\pi} = \frac{2}{\pi} \cdot \left[\frac{\pi}{2}(\pi)^2 - \frac{\pi^3}{3} - 0 \right] = \frac{\cancel{2}}{\cancel{\pi}} \cdot \frac{\pi^{\cancel{3}\,2}}{\cancel{6}\,3} = \frac{\pi^2}{3}. \Rightarrow$$

(8) $\Rightarrow \boxed{a_0 = \frac{\pi^2}{3}}$.

Similarly

$$a_n = \frac{2}{\pi}\int_0^{\pi} \underbrace{g(x)}_{=f(x) \text{ on } [0,\pi]} \cdot \cos(nx)\,dx) = \frac{2}{\pi}\cdot\int_0^{\pi} f(x)\cos(nx)\,dx =$$

$$= \frac{2}{\pi}\cdot\int_0^{\pi} x(\pi - x)\cos(nx)\,dx = \frac{2}{\pi}\left\{\int_0^{\pi}\pi x\cos(nx)\,dx - \int_0^{\pi} x^2\cos(nx)\,dx\right\}$$

$$= \frac{2}{\pi}\cdot\left\{\pi\cdot\underbrace{\int_0^{\pi} x\cdot\cos(nx)\,dx}_{I_1} - \underbrace{\int_0^{\pi} x^2\cos(nx)\,dx}_{I_2}\right\}.$$

$$I_1 = \int_0^{\pi} x\cos(nx)\,dx = \int_0^{\pi} \underset{u}{x}\; d\underbrace{\left[\frac{\sin(nx)}{n}\right]}_{v} = \overbrace{x\cdot\frac{\sin(nx)}{n}\Big|_{x=0}^{\pi}}^{=0} - \frac{1}{n}\int_0^{\pi}\sin(nx)\,dx =$$

$$= 0 - \frac{1}{n}\cdot\left[-\frac{\cos(nx)}{n}\right]_{x=0}^{\pi} = \frac{1}{n^2}\left[\cos(n\pi) - \cos(n\cdot 0)\right] = \frac{1}{n^2}\cdot\left[\cos(n\pi) - 1\right]$$

(9) $\Rightarrow I_1 = \frac{1}{n^2}\cdot\left[\cos(n\pi) - 1\right]$.

$$I_2 = \int_0^{\pi} x^2\cdot\cos(nx)\,dx = \int_0^{\pi} \underset{u}{x^2}\cdot d\underbrace{\left[\frac{\sin(nx)}{n}\right]}_{v} = x^2\cdot\overbrace{\frac{\sin(nx)}{n}\Big|_0^{\pi}}^{=0} - \int_0^{\pi}\frac{\sin(nx)}{n}\,d(x^2)$$

$$= 0 - \frac{2}{n}\int_0^{\pi} x\sin(nx)\,dx = -\frac{2}{n}\cdot\int_0^{\pi}\underset{u}{x}\; d\underbrace{\left[\frac{-\cos(nx)}{n}\right]}_{v} =$$

$$= -\frac{2}{n}\cdot\left\{x\cdot\left[-\frac{\cos(nx)}{n}\right]_{x=0}^{\pi} - \int_0^{\pi}-\frac{\cos(nx)}{n}\cdot dx\right\} =$$

$$= -\frac{2}{n}\cdot\left\{\frac{-\pi\cos(n\pi)}{n} + \frac{0\cdot\cancel{-}}{n} + \frac{1}{n}\cdot\int_0^{\pi}\cos(nx)\,dx\right\} =$$

$$= \frac{2\pi\cos(n\pi)}{n^2} - \frac{2}{n^2}\cdot\left[\overbrace{\frac{\sin(nx)}{n}}^{=0}\right]_{x=0}^{\pi} = \frac{2\pi\cdot\cos(n\pi)}{n^2}. \Rightarrow$$

(10) $I_2 = \dfrac{2\pi\cos(n\pi)}{n^2}.$

$$a_n = \frac{2}{\pi}\left\{I_1\pi - I_2\right\} = \frac{2}{\pi}\cdot\left\{\pi\cdot\left[\frac{1}{n^2}\left(\cos(n\pi)-1\right)\right] - \frac{2\pi\cos(n\pi)}{n^2}\right\}$$

$$= \frac{2}{\pi}\cdot\left\{\frac{\pi\cdot\cos(n\pi) - \pi - 2\pi\cos(n\pi)}{n^2}\right\} = \frac{2}{n^2}\left\{-1-\cos(n\pi)\right\} = \frac{-2\left(1+\cos(n\pi)\right)}{n^2}.$$

(11) $\boxed{a_n = \dfrac{-2-2\cos(\pi n)}{n^2}}$, $n = 1, 2, \ldots$

By (7),

( ) $\boxed{b_n = 0}$, $n = 1, 2, \ldots$

Thus the F-S in (4) has the form:

(12) F.S of $g(x) = \dfrac{\pi^2}{6} + \displaystyle\sum_{n=1}^{\infty}\left[\frac{-2-2\cos(\pi n)}{n^2}\right]\cos(nx)$

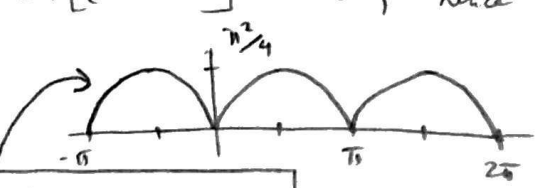
But in (12), note that if $n = 2k-1$, then we have

so let

$\boxed{n = 2k-1}$ $k = 1, 2, \ldots$, then

(13) $$+ \sum_{k=1}^{\infty} \underbrace{\frac{-2-2\overbrace{\cos[(2k-1)\pi]}^{(-1)}}{(2k-1)^2}}_{=0}\cos[(2k-1)x] = 0, \text{ hence}$$

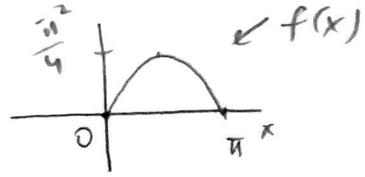
$\boxed{n = 2k}$, $k = 1, 2.$



(14) $$\boxed{\text{F.S of } g(x) = \frac{\pi^2}{6} - \sum_{k=1}\frac{1+\cos(2k\pi)}{2k^2}\cdot\cos(2kx)}$$, matches 8.33 in HW10

4.(ii) $x\cdot(\pi - x)$ is written by sines.

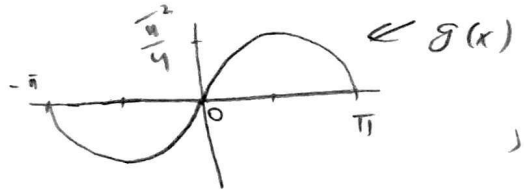
Let



(1) $f(x) = x(\pi - x)$ on $0 \le x \le \pi$.

let $g(x)$ to be an odd extension of $f(x)$ on $[-\pi, \pi]$;

(3) $$g(x) := \begin{cases} f(x), & 0 \le x \le \pi \\ -f(x), & -\pi \le x < 0 \end{cases}$$



Then F-S of odd f-n $g(x)$ on $[-\pi, \pi]$ is

(4) F-S of $g(x) = g(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos(nx) + b_n \sin(nx)\right]$

$$\boxed{\begin{array}{l} \text{(a) } g(x) \text{ is piecewise smooth} \\ \text{(b) } g(-\pi) = g(\pi) \end{array}}$$

where

(5) $a_n = 0$, $n = 0, 1, 2, ..$

(6) $b_n = \frac{2}{\pi}\int_0^{\pi} g(x)\sin(nx)\,dx$, $n = 1, 2, ..$ by Pr. 8 on HW 10.

Then

$$b_n = \frac{2}{\pi}\int_0^{\pi} \underbrace{g(x)}_{=f(x) \text{ on } [0,\pi]}\sin(nx)\,dx = \frac{2}{\pi}\cdot\int_0^{\pi} f(x)\sin(nx)\,dx =$$

$$= \frac{2}{\pi}\cdot\int_0^{\pi} x(\pi - x)\cdot\sin(nx)\,dx = \frac{2}{\pi}\cdot\left\{\pi\cdot\underbrace{\int_0^{\pi} x\sin(nx)\,dx}_{I_1} - \underbrace{\int_0^{\pi} x^2\cdot\sin(nx)\,dx}_{I_2}\right\} \Rightarrow$$

(7) $b_n = \frac{2}{\pi}\left\{\pi\cdot I_1 - I_2\right\}$ where

$$I_1 = \int_0^{\pi} x \cdot \sin(nx)\,dx = \int_0^{\pi} \underbrace{x}_{u} \cdot d\underbrace{\left[\frac{-\cos(nx)}{n}\right]}_{v} = x \cdot \left[\frac{-\cos(nx)}{n}\right]_{x=0}^{\pi} - \int_0^{\pi} \frac{-\cos(nx)}{n}\,dx$$

$$= -\left[\frac{\pi\cos(n\pi)}{n} - 0\right] + \frac{1}{n}\int_0^{\pi} \cos(nx)\,dx = \frac{-\pi\cos(n\pi)}{n} + \frac{1}{n}\cdot\left[\frac{\sin(nx)}{n}\right]_{x=0}^{\pi}$$

$$= \frac{-\pi\cos(n\pi)}{n} + \frac{1}{n^2}\cdot[0-0] = \frac{-\pi\cdot\cos(n\pi)}{n}. \quad \Rightarrow$$

(8) $$I_1 = \frac{-\pi\cdot\cos(n\pi)}{n},$$

$$I_2 = \int_0^{\pi} x^2 \cdot \sin(nx)\,dx = \int_0^{\pi} \underbrace{x^2}_{u} \cdot d\underbrace{\left[\frac{-\cos(nx)}{n}\right]}_{v} = x^2 \cdot \left[\frac{-\cos(nx)}{n}\right]_{x=0}^{\pi} -$$

$$- \int_0^{\pi} -\frac{\cos(nx)}{n}\,2x\,dx = \frac{-\pi^2\cdot\cos(n\pi)}{n} + \frac{2}{n}\int_0^{\pi} x\cos(nx)\,dx =$$

$$= \frac{-\pi^2\cdot\cos(n\pi)}{n} + \frac{2}{n}\cdot\int_0^{\pi} \underbrace{x}_{u}\, d\underbrace{\left[\frac{\sin(nx)}{n}\right]}_{v} = \frac{-\pi^2\cdot\cos(n\pi)}{n} +$$

$$+ \frac{2}{n}\cdot\left\{ x\cdot\frac{\sin(nx)}{n}\Big|_{x=0}^{\pi} - \int_0^{\pi} \frac{\sin(nx)}{n}\,dx \right\} = \frac{-\pi^2\cdot\cos(n\pi)}{n} +$$

$$+ \frac{2}{n}\left\{ -\frac{1}{n}\cdot\left[\frac{-\cos(nx)}{n}\right]_{x=0}^{\pi} \right\} = \frac{-\pi^2\cdot\cos(n\pi)}{n} + \frac{2}{n^3}\left[\cos(n\pi) - 1\right].$$

(9) $$I_2 = \frac{-\pi^2\cdot\cos(n\pi)}{n} + \frac{2\cdot(\cos(n\pi)-1)}{n^3}.$$

$$b_n = \frac{2}{\pi}\cdot\left\{\pi\cdot I_1 - I_2\right\} = \frac{2}{\pi}\cdot\left\{\pi\cdot\left[\frac{-\pi\cos(n\pi)}{n}\right] + \frac{\pi^2\cos(n\pi)}{n} - \frac{2[\cos(n\pi)-1]}{n^3}\right\}$$

$$\Rightarrow$$

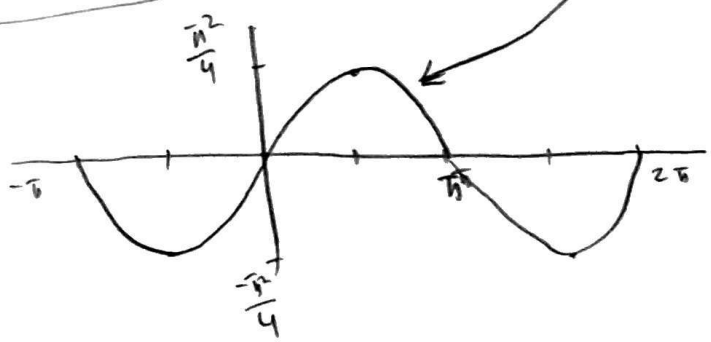
(10) $\boxed{b_n = \frac{-4}{\pi n^3} \cdot [\cos(n\pi) - 1]}$ $n = 1, 2 \ldots$. Then (4) is

(11) F-S of $g(x) = g(x) = \sum_{n=1}^{\infty} b_n \sin(nx) =$

$$= \sum_{n=1}^{\infty} \frac{-4}{\pi n^3} [\cos(\pi n) - 1] \cdot \sin(nx).$$

Note in (11) if $n = 2k \Rightarrow \sum \square \overbrace{(1-1)}^{\infty} \square = 0$, so

let $\boxed{n = 2k-1}$ for $k = 1, 2 \ldots$ Then (11) is

(12) $$\boxed{g(x) = -\frac{4}{\pi} \sum_{k=1}^{\infty} \frac{1}{(2k-1)^3} \sin[(2k-1)x]}$$

matching (8.33) in Hw 10 ✓